

Sam Ethan Mathew

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Summary

AI/Robotics-oriented CMU Engineering student focusing on motion planning, perception, and systems (ROS2). Experience with A* and heuristic search, CV pipelines (YOLO/OpenCV), simulation (Gazebo/Isaac), and production web tooling (Next.js+Supabase). Seeking roles in AI, Robotics, or SWE.

Education

Carnegie Mellon University, College of Engineering — B.S. Candidate Aug 2025 – May 2028 (expected)

- Relevant coursework (in progress/planned): Discrete Math (21-127), Matrices & Linear Transformations (21-241), Intro ECE (18-100), Data Structures (15-122), Probability (21-325).
- Dual enrollment (De Anza College): Data Abstraction & Structures (CIS 22C), C++ Programming (CIS 22A/22B), Python Programming (CIS 41A).

Skills

Programming: Python (NumPy, OpenCV, basic PyTorch/TensorFlow), C++, Java, TypeScript **Robotics:** ROS2/rcldpy, URDF/RViz/Gazebo, A* & heuristic search **ML/CV:** YOLOv8, scikit-learn, ONNX export, real-time inference **Systems/Tools:** Git/GitHub, Linux/WSL, Docker, Supabase, Next.js, Tailwind, shadcn/ui

Experience

Student Researcher, CMU BioRobotics Lab (MT-TSP / Path Planning) — Pittsburgh, PA Sept 2025 – Present

- Implement A* and heuristic variants; benchmark against NetworkX to verify optimality and runtime baselines.
- Prototype moving-target path planning in polygonal environments; explore Polyanya-style search for dynamic goals.
- Build reproducible experiments in Python (NumPy) with evaluation harnesses and simple visualizers (RViz/2D grid).

Software Developer, ScottyLabs — Universify Platform — Pittsburgh, PA Sept 2025 – Present

- Contribute to a real-time student activities platform (pickup sports, games) with join/RSVP and moderation workflows.
- Stack: React.js (App Router) + TypeScript + Supabase (Auth/DB/Storage) + edge APIs; implement event feeds and geofilters.
- Built and deployed a recommendation system using collaborative filtering and user profile matching to personalize activity suggestions for students.

Selected Projects

BizGrow — Gamified AI Micro-Loan Incubator (HackCMU) Sept 2025

- Designed a quests → XP/coins → eligibility funnel with deterministic scoring and admin review tools.
- Built Next.js + Tailwind + shadcn/ui front-end; Supabase Postgres/Auth/Storage; endpoints for quest review and eligibility.

Speech → ROS2 Motion Pipeline (SoloTech Open Source) Summer 2025

- Converted free-form speech to structured JSON → ROS2 actions; validated multi-command sequences in RViz/TurtleBot.
- Generalized to SO-100 arm; hardened URDF loading and command validation for safer execution.

Helmet Compliance Detector (YOLOv8 + OpenCV) 2024

- Curated dataset and trained/exported ONNX; achieved ~35 FPS real-time stream with timestamped detections.

Awards & Activities

- Founding SW Mentor, RoboRoar FTC 30449 (2025–Present); FRC 2473 “Gold Strikers” Alumni/Electrical Lead
- FIRST Robotics "Excellence in Engineering" (2025)
- FBLA State: 2nd (Game & Simulation 2025, Web Dev 2024); Chapter President (250+ members, \$30k+ raised)